

A Canonical Master Equation for Swirl–String Theory Unified Mass, Clock, Geometric Gate, and Topological Kernel

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Abstract

We derive a compact master equation for Swirl–String Theory (SST) that absorbs the currently separated sectors of mass generation, geometric shielding, swirl-clock kinematics, and topological state selection into a single hierarchical formula. The resulting expression factors into four pieces: a medium scale, a geometric gate, a topological kernel, and a local clock-impedance factor. We show that the formula is dimensionally exact, reduces to the existing SST mass-functional and swirl-clock sectors in the appropriate limits, and numerically reproduces the canonical geometric gate $\lambda_c/(\pi r_c) = 4/\alpha$ using the canonical SST constants. The result is proposed as a canonical organizing equation rather than as a completed full derivation of all coefficient values.

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1 Purpose

Several SST papers presently encode closely related structures in different notations:

- mass as localized swirl energy,
- the Swirl-Clock as a local slowing factor,
- the shielding gate as the ratio $\lambda_c/(\pi r_c)$,
- topological state dependence through crossing, ropelength, helicity, and Golden-layer factors.

The aim of this article is to replace those parallel formulations by one canonical equation.

2 Primitive definitions

We take as primitive:

$$\rho_E(x) = \frac{1}{2}\rho_f(x) \|\mathbf{v}_\odot(x)\|^2, \quad \rho_m(x) = \frac{\rho_E(x)}{c^2}, \quad (1)$$

so that

$$\rho_m(x) = \frac{1}{2}\rho_f(x) \frac{\|\mathbf{v}_\odot(x)\|^2}{c^2}. \quad (2)$$

The local matter-branch swirl clock is

$$S_t^\odot(x) = \sqrt{1 - \frac{\|\mathbf{v}_\odot(x)\|^2}{c^2}}. \quad (3)$$

We also define the binary geometric exposure gate

$$G(K) \in \{0, 1\}, \quad (4)$$

Definition 1 (Topological kernel). *The canonical dimensionless topology kernel is taken to be*

$$\Xi_K = [\alpha_C C(K) + \beta_L L(K) + \gamma_H \mathcal{H}(K)] \varphi^{-2k(K)}, \quad (5)$$

where $C(K)$ is a crossing-type invariant, $L(K)$ a ropelength-type invariant, $\mathcal{H}(K)$ a helicity/Hopf-type invariant contribution, and $k(K)$ the discrete Golden-layer index.

3 Geometric gate lemma

Lemma 1 (Canonical geometric shielding gate). *The geometric sector factor is*

$$\Pi_K = \left(\frac{\lambda_c}{\pi r_c} \right)^{G(K)}. \quad (6)$$

Using the SST closure relation, this is equivalently

$$\Pi_K = \left(\frac{4}{\alpha} \right)^{G(K)}. \quad (7)$$

Proof. The existing SST geometric closure identifies the dimensionless shielding ratio as

$$\frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha}. \quad (8)$$

Since $G(K)$ is binary, the state either remains unamplified ($G = 0$) or receives one geometric exposure factor ($G = 1$). $\square$

4 Master equation

Theorem 1 (Canonical SST master equation). *For a topological state K and local position x , define*

$$M_K(x) = \rho_m(x) r_c^3 \Pi_K \Xi_K S_t^\circ(x)^{-2}. \quad (9)$$

Equivalently,

$$M_K(x) = \left(\frac{\rho_f(x) \|\mathbf{v}_\circ(x)\|^2}{2c^2} \right) r_c^3 \left(\frac{\lambda_c}{\pi r_c} \right)^{G(K)} [\alpha_C C(K) + \beta_L L(K) + \gamma_H \mathcal{H}(K)] \varphi^{-2k(K)} S_t^\circ(x)^{-2} \quad (10)$$

with

$$S_t^\circ(x) = \sqrt{1 - \frac{\|\mathbf{v}_\circ(x)\|^2}{c^2}}. \quad (11)$$

Dimensional check

We verify the units of the prefactor:

$$\left[\frac{\rho_f \|\mathbf{v}_\circ\|^2}{c^2} r_c^3 \right] = \left[\frac{\text{kg m}^{-3} \cdot \text{m}^2 \text{s}^{-2}}{\text{m}^2 \text{s}^{-2}} \cdot \text{m}^3 \right] = \text{kg}. \quad (12)$$

All remaining factors are dimensionless. Therefore M_K has units of mass.

5 Reduction limits

5.1 Static spectrum limit

If the local state is approximately stationary with

$$\rho_f(x) \rightarrow \rho_f, \quad \|\mathbf{v}_\odot(x)\| \rightarrow \|\mathbf{v}_\odot\|, \quad S_t^\odot(x) \rightarrow 1, \quad (13)$$

then

$$M_K = \left(\frac{\rho_f \|\mathbf{v}_\odot\|^2}{2c^2} \right) r_c^3 \left(\frac{\lambda_c}{\pi r_c} \right)^{G(K)} \Xi_K. \quad (14)$$

This is the canonical static mass-spectrum form.

5.2 Geometric-mass limit

If one packages the topological kernel into a ropelength/topology factor

$$\Xi_K \longrightarrow \tilde{L}_{\text{tot}}(K) \mathcal{K}(K), \quad (15)$$

then

$$M_K = \rho_m r_c^3 \left(\frac{\lambda_c}{\pi r_c} \right)^{G(K)} \tilde{L}_{\text{tot}}(K) \mathcal{K}(K), \quad (16)$$

which is the pure geometric mass-functional structure.

5.3 Clock-only limit

If the state label K is fixed and only the local swirl field varies, then

$$M_K(x) \propto S_t^\odot(x)^{-2}. \quad (17)$$

Thus the same local swirl field that slows the clock also strengthens the local effective mass scale.

6 Weak-swirl expansion

Since

$$S_t^{\odot-2}(x) = \frac{1}{1 - \|\mathbf{v}_\odot(x)\|^2/c^2}, \quad (18)$$

one has for $\|\mathbf{v}_\odot\|^2/c^2 \ll 1$,

$$S_t^{\odot-2}(x) = 1 + \frac{\|\mathbf{v}_\odot(x)\|^2}{c^2} + \mathcal{O}\left(\frac{\|\mathbf{v}_\odot\|^4}{c^4}\right). \quad (19)$$

Hence the leading correction is positive and quadratic in the local swirl speed.

7 Numerical benchmark using canonical SST constants

Use

$$\|\mathbf{v}_\odot\| = 1.09384563 \times 10^6 \text{ m s}^{-1}, \quad r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad \rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3}. \quad (20)$$

Then

$$\rho_E = \frac{1}{2} \rho_f \|\mathbf{v}_\odot\|^2 \approx 4.1877439 \times 10^5 \text{ J m}^{-3}, \quad (21)$$

and

$$\rho_m = \frac{\rho_E}{c^2} \approx 4.6594935 \times 10^{-12} \text{ kg m}^{-3}. \quad (22)$$

The bare mass scale is

$$M_\star = \rho_m r_c^3 \approx 1.303 \times 10^{-56} \text{ kg}. \quad (23)$$

The geometric gate is

$$\frac{\lambda_c}{\pi r_c} \approx 5.48144 \times 10^2, \quad (24)$$

which matches

$$\frac{4}{\alpha} \approx 5.48144 \times 10^2. \quad (25)$$

The Swirl-Clock at the canonical swirl speed is

$$S_t^\circ = \sqrt{1 - \frac{\|\mathbf{v}_\circ\|^2}{c^2}} \approx 0.999993344, \quad (26)$$

hence

$$S_t^{\circ-2} \approx 1.0000133. \quad (27)$$

Therefore:

1. the absolute mass scale is set by $\rho_f, \mathbf{v}_\circ, r_c$,
2. the hierarchy must predominantly come from $\Pi_K \Xi_K$,
3. the clock factor is a small but coherent impedance correction in the canonical regime.

8 Interpretation

The master equation separates SST into four logically distinct sectors:

$$\boxed{\text{medium scale} \times \text{geometric gate} \times \text{topological kernel} \times \text{clock impedance}} \quad (28)$$

More explicitly,

$$M_K(x) \sim (\text{localized swirl energy density}) \times (\text{core volume}) \times (\text{state geometry}) \times (\text{topological identity}) \times (\text{local time}) \quad (29)$$

9 Status and assumptions

Derived / structurally fixed

$$\rho_E = \frac{1}{2} \rho_f \|\mathbf{v}_\circ\|^2, \quad \rho_m = \rho_E / c^2, \quad S_t^\circ = \sqrt{1 - \|\mathbf{v}_\circ\|^2 / c^2}, \quad \frac{\lambda_c}{\pi r_c} = \frac{4}{\alpha}. \quad (30)$$

Closure-level / modeled

$$G(K) \in \{0, 1\}, \quad \Xi_K = [\alpha_C C + \beta_L L + \gamma_H \mathcal{H}] \varphi^{-2k}. \quad (31)$$

Open

The exact coefficient values $\alpha_C, \beta_L, \gamma_H$, the optimal invariant set, and whether the clock coupling is best represented by $S_t^{\circ-2}$, $S_t^{\circ-1}$, or a weaker deformation ansatz.

10 Falsifiable consequences

1. If two states have identical Π_K and Ξ_K , then their rest masses must differ only through local medium variations.
2. If the geometric gate interpretation is correct, state families should cluster by binary exposure class $G(K)$.
3. If the clock-impedance coupling is correct, strong local swirl regions should induce a correlated increase in effective inertial scale and proper-time slowing.

A kid analogy: a particle is like a tied spinning knot in water; its mass comes from how much spinning is packed into the knot, how complicated the knot is, and how much the local spinning slows the knot's own rhythm.

References

- [1] G. K. Batchelor, 1967, *An Introduction to Fluid Dynamics*, Cambridge University Press, permalink: <https://doi.org/10.1017/CB09780511800955>.
- [2] P. G. Saffman, 1992, *Vortex Dynamics*, Cambridge University Press, permalink: <https://doi.org/10.1017/CB09780511624063>.
- [3] L. Allen, M. W. Beijersbergen, R. J. C. Spreeuw, and J. P. Woerdman, 1992, *Orbital angular momentum of light and the transformation of Laguerre-Gaussian laser modes*, Physical Review A **45**, 8185–8189, DOI: <https://doi.org/10.1103/PhysRevA.45.8185>.
- [4] T. Jacobson and D. Mattingly, 2004, *Einstein-Aether gravity: a status report*, Foundations of Physics **43**, permalink: <https://doi.org/10.1007/s10701-012-9661-5>.
- [5] B. P. Abbott et al., 2017, *GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral*, Physical Review Letters **119**, 161101, DOI: <https://doi.org/10.1103/PhysRevLett.119.161101>.
- [6] O. Iskandarani, 2026, *Swirl-String Theory Canon and related SST papers*, Independent Research / project canon.